CPSC 304 Project Cover Page

Milestone #: 4

Date: Apr 2, 2026

Group Number: 113

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

1. SQL script

You can find the sql script at
https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/sql/setup.sql

2. Description

2.1 Summary of the project

This app helps beginners save money by tracking their groceries. It uses AI to suggest recipes based on what's in their kitchen, and users can share their inventory across multiple accounts. Users can also share their successful meals so the community can find recipes that actually work.

2.2 Description of the final project

We've finished all planned features and also added some new features to satisfy the queries requirement.

2.3 Final schema vs the schema we turned in

We didn't change the schema in milestone 4

2.4 Visual representation of the state of the database after running the .sql

Account:

account

accountid	email	username	password
11111111-1111-1111-1111-111111111111	user1@email.com	Joseph	hashedpass1
22222222-2222-2222-2222-222222222222	user2@email.com	Ali	hashedpass2
33333333-3333-3333-3333-333333333333	user3@email.com	Runjuu	hashedpass3
44444444-4444-4444-4444-444444444444	user4@email.com	Ivan	hashedpass4
55555555-5555-5555-5555-555555555555	user5@email.com	Bob	hashedpass5
66666666-6666-6666-6666-666666666666	user6@email.com	Rob	hashedpass6
77777777-7777-7777-7777-777777777777	user7@email.com	Jane	hashedpass7
88888888-8888-8888-8888-888888888888	user8@email.com	John	hashedpass8
99999999-9999-9999-9999-999999999999	user9@email.com	Cinda	hashedpass9
aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa	user10@email.com	Gregor	hashedpass10

TrialAccount:

trialaccount

accountid	usageremaining	trialstartdate	trialenddate
11111111-1111-1111-1111-111111111111	10	2026-03-01 02:00:00-08	2026-03-15 03:00:00-07
22222222-2222-2222-2222-222222222222	5	2026-03-02 03:30:00-08	2026-03-16 04:30:00-07
33333333-3333-3333-3333-333333333333	0	2026-02-01 01:00:00-08	2026-02-15 01:00:00-08
44444444-4444-4444-4444-444444444444	15	2026-03-05 00:00:00-08	2026-03-19 01:00:00-07
55555555-5555-5555-5555-555555555555	2	2026-03-04 06:00:00-08	2026-03-18 07:00:00-07

PaidAccount:

paidaccount

accountid	subscriptionstartdate	subscriptionenddate	subscriptiontype
66666666-6666-6666-6666-666666666666	2026-01-01 02:00:00-08	2026-02-01 02:00:00-08	monthly
77777777-7777-7777-7777-777777777777	2026-01-15 02:00:00-08	2027-01-15 02:00:00-08	yearly
88888888-8888-8888-8888-888888888888	2026-02-01 02:00:00-08	2026-03-01 02:00:00-08	monthly
99999999-9999-9999-9999-999999999999	2025-11-01 03:00:00-07	2026-11-01 02:00:00-08	yearly
aaaaaaaa-aaaa-aaaa-aaaa-aaaaaaaaaaaa	2026-03-01 02:00:00-08	2026-04-01 03:00:00-07	monthly

EmojiMeaning:

emojimeaning

emoji	kind
👍	Positive
👎	Negative
🔥	Positive
😐	Neutral
😡	Negative

Ingredient:

ingredient

name	carbs	protein	fat
Broccoli	6.60	2.80	0.40
Chicken Breast	0.00	31.00	3.60
Egg	1.10	6.30	5.30
Olive Oil	0.00	0.00	14.00
White Rice	28.00	2.70	0.30

Inventory:

inventory

inventoryid	name	description	type
d1111111-1111-1111-1111-111111111111	Main Fridge	Kitchen refrigerator	Fridge
d2222222-2222-2222-2222-222222222222	Pantry	Dry goods storage	Pantry
d3333333-3333-3333-3333-333333333333	Basement Freezer	Deep freeze meats	Freezer
d4444444-4444-4444-4444-444444444444	Spice Rack	Spices and herbs	Rack
d5555555-5555-5555-5555-555555555555	Office Mini Fridge	Snacks for work	Fridge

InventoryItem:

inventoryitem

inventoryitemid	inventoryid	quantity	creationdate	expirationdate	ingredientname	receiptid
1	d1111111-1111-1111-11…	5	2026-03-01 01:30:00-08	2026-03-10 02:30:00-07	Chicken Breast	c1111111-1111-1111-1…
1	d2222222-2222-2222-22…	12	2026-03-02 01:30:00-08	2026-04-02 02:30:00-07	Egg	c2222222-2222-2222-2…
1	d3333333-3333-3333-33…	1	2026-03-03 01:30:00-08	NULL	Olive Oil	c3333333-3333-3333-3…
1	d4444444-4444-4444-44…	3	2026-03-04 01:30:00-08	2026-03-11 02:30:00-07	Broccoli	c4444444-4444-4444-4…
1	d5555555-5555-5555-55…	2	2026-03-05 01:30:00-08	2027-03-05 01:30:00-08	White Rice	c5555555-5555-5555-5…
2	d1111111-1111-1111-11…	2	2026-03-01 01:35:00-08	2026-03-20 02:35:00-07	Broccoli	c1111111-1111-1111-1…
2	d2222222-2222-2222-22…	1	2026-03-02 01:35:00-08	NULL	Olive Oil	c2222222-2222-2222-2…
2	d3333333-3333-3333-33…	4	2026-03-03 01:35:00-08	2026-03-18 02:35:00-07	Chicken Breast	NULL
2	d4444444-4444-4444-44…	6	2026-03-04 01:35:00-08	2026-06-01 02:35:00-07	Egg	NULL
2	d5555555-5555-5555-55…	3	2026-03-05 01:35:00-08	2026-03-22 02:35:00-07	Egg	NULL
3	d1111111-1111-1111-11…	1	2026-03-01 01:40:00-08	2026-04-15 02:40:00-07	White Rice	c1111111-1111-1111-1…
3	d2222222-2222-2222-22…	2	2026-03-02 01:40:00-08	2026-03-16 02:40:00-07	Broccoli	NULL
3	d5555555-5555-5555-55…	1	2026-03-05 01:40:00-08	NULL	Olive Oil	NULL

AccountOwnsInventory:

accountownsinventory

accountid	inventoryid
11111111-1111-1111-1111-111111111111	d1111111-1111-1111-1111-111111111111
11111111-1111-1111-1111-111111111111	d2222222-2222-2222-2222-222222222222
22222222-2222-2222-2222-222222222222	d1111111-1111-1111-1111-111111111111
22222222-2222-2222-2222-222222222222	d2222222-2222-2222-2222-222222222222
33333333-3333-3333-3333-333333333333	d1111111-1111-1111-1111-111111111111
66666666-6666-6666-6666-666666666666	d3333333-3333-3333-3333-333333333333
77777777-7777-7777-7777-777777777777	d3333333-3333-3333-3333-333333333333
77777777-7777-7777-7777-777777777777	d4444444-4444-4444-4444-444444444444
88888888-8888-8888-8888-888888888888	d4444444-4444-4444-4444-444444444444
88888888-8888-8888-8888-888888888888	d5555555-5555-5555-5555-555555555555
99999999-9999-9999-9999-999999999999	d5555555-5555-5555-5555-555555555555

Post:

post

postid	title	body	creationdate	visibility	accountid	recipeid
e1111111-1111-1111-1111…	My First Meal	Turned out great!	2026-03-01 04:00:00-08	public	11111111-1111-1111…	b1111111-1111-1111-111…
e2222222-2222-2222-2222…	Breakfast Success	Love these eggs.	2026-03-02 04:00:00-08	public	22222222-2222-2222…	b2222222-2222-2222-222…
e3333333-3333-3333-3333…	Healthy Eating	Broccoli is the best.	2026-03-03 04:00:00-08	private	66666666-6666-6666…	b3333333-3333-3333-333…
e4444444-4444-4444-4444…	Dinner Party	Made stir fry for friends.	2026-03-04 04:00:00-08	public	77777777-7777-7777…	b4444444-4444-4444-444…
e5555555-5555-5555-5555…	Quick Lunch	Fried rice in 15 mins.	2026-03-05 04:00:00-08	public	88888888-8888-8888…	b5555555-5555-5555-555…

PostReaction:

postreaction

postid	accountid	emoji
e1111111-1111-1111-1111-111111111111	22222222-2222-2222-2222-222222222222	👍
e2222222-2222-2222-2222-222222222222	66666666-6666-6666-6666-666666666666	🔥
e3333333-3333-3333-3333-333333333333	77777777-7777-7777-7777-777777777777	👍
e4444444-4444-4444-4444-444444444444	88888888-8888-8888-8888-888888888888	😐
e5555555-5555-5555-5555-555555555555	11111111-1111-1111-1111-111111111111	🔥

Receipt:

receipt

receiptid	ocrresult	imagefilepath	creationdate	accountid
c1111111-1111-1111-1111-111111111111	Chicken, Rice	/images/receipts/1.jpg	2026-03-01 01:00:00-08	11111111-1111-1111-11…
c2222222-2222-2222-2222-222222222222	Eggs, Broccoli	/images/receipts/2.jpg	2026-03-02 01:00:00-08	22222222-2222-2222-22…
c3333333-3333-3333-3333-333333333333	Olive Oil	/images/receipts/3.jpg	2026-03-03 01:00:00-08	66666666-6666-6666-66…
c4444444-4444-4444-4444-444444444444	Chicken, Eggs	/images/receipts/4.jpg	2026-03-04 01:00:00-08	77777777-7777-7777-77…
c5555555-5555-5555-5555-555555555555	Rice, Broccoli	/images/receipts/5.jpg	2026-03-05 01:00:00-08	88888888-8888-8888-88…

Recipe:

recipe

recipeid	name	content	costincents	time	cuisine	creationdate	modificationdate	accountid
b1111111-1111-11…	Chicken & Rice	Grill chicken, boil…	450	30	American	2026-03-01…	2026-03-01 02…	11111111-1111-1111-111…
b1111111-1111-11…	Chicken Broccol…	Roast chicken, stea…	525	20	American	2026-03-31…	2026-03-31 03…	11111111-1111-1111-111…
b1111111-1111-11…	Chicken Broccol…	Saute chicken and b…	575	20	American	2026-03-31…	2026-03-31 03…	11111111-1111-1111-111…
b1111111-1111-11…	Fried Eggs	Fry eggs in pan.	150	5	American	2026-03-02…	2026-03-02 02…	11111111-1111-1111-111…
b1111111-1111-11…	Steamed Broccoli	Steam until tender.	200	10	American	2026-03-03…	2026-03-03 02…	11111111-1111-1111-111…
b1111111-1111-11…	Chicken Stir Fry	Fry chicken and bro…	600	20	Asian	2026-03-04…	2026-03-04 02…	11111111-1111-1111-111…
b1111111-1111-11…	Egg Fried Rice	Mix egg, rice, and…	300	20	Asian	2026-03-05…	2026-03-05 02…	11111111-1111-1111-111…
b2222222-2222-22…	Fried Eggs	Fry eggs in pan.	150	5	American	2026-03-02…	2026-03-02 02…	22222222-2222-2222-222…
b3333333-3333-33…	Steamed Broccoli	Steam until tender.	200	10	American	2026-03-03…	2026-03-03 02…	66666666-6666-6666-666…
b4444444-4444-44…	Chicken Stir Fry	Fry chicken and bro…	600	20	Asian	2026-03-04…	2026-03-04 02…	77777777-7777-7777-777…
b5555555-5555-55…	Egg Fried Rice	Mix egg, rice, and…	300	20	Asian	2026-03-05…	2026-03-05 02…	88888888-8888-8888-888…

RecipeNote:

recipenote

recipenoteid	photo	note	creationdate	modificationdate	accountid
f1111111-1111-1111-11…	NULL	Added extra salt.	2026-03-01 03:00:00-08	2026-03-01 03:00:00-08	11111111-1111-1111-111…
f2222222-2222-2222-22…	NULL	Burned the eggs a bit.	2026-03-02 03:00:00-08	2026-03-02 03:00:00-08	22222222-2222-2222-222…
f3333333-3333-3333-33…	NULL	Perfect texture.	2026-03-03 03:00:00-08	2026-03-03 03:00:00-08	66666666-6666-6666-666…
f4444444-4444-4444-44…	NULL	Used less oil next time.	2026-03-04 03:00:00-08	2026-03-04 03:00:00-08	77777777-7777-7777-777…
f5555555-5555-5555-55…	NULL	Family loved it!	2026-03-05 03:00:00-08	2026-03-05 03:00:00-08	88888888-8888-8888-888…

RecipeNoteAnnotatesRecipe:

recipenoteannotatesrecipe

recipenoteid	recipeid
f1111111-1111-1111-1111-111111111111	b1111111-1111-1111-1111-111111111111
f2222222-2222-2222-2222-222222222222	b2222222-2222-2222-2222-222222222222
f3333333-3333-3333-3333-333333333333	b3333333-3333-3333-3333-333333333333
f4444444-4444-4444-4444-444444444444	b4444444-4444-4444-4444-444444444444
f5555555-5555-5555-5555-555555555555	b5555555-5555-5555-5555-555555555555

Subscription:

subscription

subscriptiontype	subscriptionpriceincents
monthly	999
yearly	9999

RecipeIngredient:

recipeingredient

recipeid	ingredientname	quantity
b1111111-1111-1111-1111-111111111111	Broccoli	1
b1111111-1111-1111-1111-111111111111	Chicken Breast	2
b1111111-1111-1111-1111-111111111111	Olive Oil	1
b1111111-1111-1111-1111-111111111111	White Rice	1
b1111111-1111-1111-1111-111111111112	Broccoli	2
b1111111-1111-1111-1111-111111111112	Chicken Breast	2
b1111111-1111-1111-1111-111111111112	Olive Oil	1
b1111111-1111-1111-1111-111111111113	Broccoli	2
b1111111-1111-1111-1111-111111111113	Chicken Breast	1
b1111111-1111-1111-1111-111111111113	Egg	2
b1111111-1111-1111-1111-111111111113	Olive Oil	1
b1111111-1111-1111-1111-111111111114	Egg	2
b1111111-1111-1111-1111-111111111114	Olive Oil	1
b1111111-1111-1111-1111-111111111115	Broccoli	1
b1111111-1111-1111-1111-111111111115	Olive Oil	1
b1111111-1111-1111-1111-111111111116	Broccoli	2
b1111111-1111-1111-1111-111111111116	Chicken Breast	1
b1111111-1111-1111-1111-111111111116	Olive Oil	1
b1111111-1111-1111-1111-111111111116	White Rice	1
b1111111-1111-1111-1111-111111111117	Egg	2
b1111111-1111-1111-1111-111111111117	Olive Oil	1
b1111111-1111-1111-1111-111111111117	White Rice	1
b2222222-2222-2222-2222-222222222222	Egg	2
b2222222-2222-2222-2222-222222222222	Olive Oil	1
b3333333-3333-3333-3333-333333333333	Broccoli	1
b3333333-3333-3333-3333-333333333333	Olive Oil	1
b4444444-4444-4444-4444-444444444444	Broccoli	2
b4444444-4444-4444-4444-444444444444	Chicken Breast	1
b4444444-4444-4444-4444-444444444444	Olive Oil	1
b4444444-4444-4444-4444-444444444444	White Rice	1
b5555555-5555-5555-5555-555555555555	Egg	2
b5555555-5555-5555-5555-555555555555	Olive Oil	1
b5555555-5555-5555-5555-555555555555	White Rice	1

SQL queries

1. Insert

- file name: ./packages/backend/src/inventoryItems.service.ts
- line numbers: 73-98
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/inventoryItems.service.ts#L73-L98

2. Update

- file name: ./packages/backend/src/inventoryItems.service.ts
- line numbers: 116-125
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/inventoryItems.service.ts#L116-L125

3. Delete

- file name: ./packages/backend/src/recipes.service.ts
- line numbers: 232-233
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/recipes.service.ts#L232-L233
- This will also delete the corresponding RecipeIngredient and RecipeNoteAnnotatesRecipe rows as well

4. Selection

- file name: ./packages/backend/src/inventoryItems.service.ts
- line numbers: 19-30
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/inventoryItems.service.ts#L19-L30

5. Projection

- file name: ./packages/backend/src/recipe-notes.service.ts:
- line numbers: 112-115
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/recipe-notes.service.ts#L112-L115

6. Join

- file name: packages/backend/src/community-post.service.ts
- line numbers: 65-79
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/community-post.service.ts#L65-L79

7. Aggregation with GROUP BY

- file name: ./packages/backend/src/recipes.service.ts
- line numbers: 209-216
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/recipes.service.ts#L209-L216
- What this SQL does: it counts the ingredient number for each recipe for the given user.

```
SELECT
        r.RecipeID AS recipe_id,
        r.Name AS recipe_name,
        COUNT(*) AS ingredient_count
FROM Recipe r
JOIN RecipeIngredient ri ON ri.RecipeID = r.RecipeID
WHERE r.AccountID = ${account_id}
GROUP BY r.RecipeID, r.Name;
```

8. Aggregation with HAVING

- file name: ./packages/backend/src/recipes.service.ts:
- line numbers: 242-263
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/recipes.service.ts#L242-L263
- What this SQL does: It finds the current user's recipes that have protein levels higher than the given target.

```
SELECT
        r.RecipeID AS recipe_id,
        r.Name AS name,
        r.Content AS content,
        r.Cuisine AS cuisine,
        r.Time AS time,
        r.CostInCents AS cost_in_cents,
        r.CreationDate AS creation_date,
        r.ModificationDate AS modification_date,
        SUM(ri.Quantity * i.Protein) AS total_protein
FROM
        Recipe r,
        RecipeIngredient ri,
        Ingredient i
WHERE
        r.RecipeID = ri.RecipeID
        AND r.AccountID = ${account_id}
        AND ri.IngredientName = i.Name
GROUP BY
        r.RecipeID
HAVING
        SUM(ri.Quantity * i.Protein) >= ${target};
```

9. Nested aggregation with GROUP BY

- file name: ./packages/backend/src/ingredients.service.ts:
- line numbers: 92-115
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/ingredients.service.ts#L92-L115
- What this SQL does: It finds which ingredients appear more often than usual across all recipes. It first counts appearances for each ingredient, computes the average of those counts, and then keeps only the ingredients above that average.

```
SELECT
	ri.IngredientName AS name,
	COUNT(*) AS COUNT
FROM
	RecipeIngredient ri
GROUP BY
	ri.IngredientName
HAVING
	COUNT(*) > (
		SELECT
			AVG(COUNT)
		FROM
			(
				SELECT
					COUNT(*) AS COUNT
				FROM
					RecipeIngredient
				GROUP BY
					IngredientName
			)
	)
ORDER BY
	COUNT DESC,
	ri.IngredientName ASC;
```

10. Division

- file name: ./packages/backend/src/recipes.service.ts
- line numbers: 37-55
- https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/recipes.service.ts#L37-L55
- What this SQL does: It finds ingredients that are used in every recipe of the given user. It first checks if the user has at least one recipe, and then uses the double NOT EXISTS approach to do the division.

```
SELECT
	i.Name AS ingredient_name
FROM
	Ingredient i
WHERE
	EXISTS (
		SELECT
			*
		FROM
			Recipe r
		WHERE
			r.AccountID = ${account_id}
	)
	AND NOT EXISTS (
		SELECT
			*
		FROM
			Recipe r
		WHERE
			r.AccountID = ${account_id}
			AND NOT EXISTS (
				SELECT
					*
				FROM
					RecipeIngredient ri
				WHERE
					ri.RecipeID = r.RecipeID
					AND ri.IngredientName = i.Name
			)
	);
```

4. Use of AI tools

We didn't use AI for this milestone, but we integrated some AI features into our project.

- The user can use Gemini to generate recipe based on what they have in their inventories
    - The prompt can be found on https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/ai-recipe.service.ts#L64-L70
- The user can use Gemini to add items to inventory by uploading an image of the receipt
    - The prompt can be found on https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/receipt.service.ts#L197-L203
- The user can add ingredients by providing just the name and use Gemini to auto fill in the nutrition information, like fat, protein, and carbs
    - The prompt can be found on https://github.students.cs.ubc.ca/CPSC304-2025W-T2/team_113/blob/main/packages/backend/src/ingredients.service.ts#L197-L202